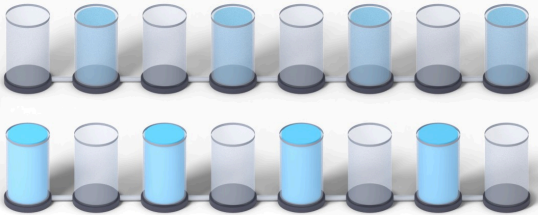
(a) $V = 0$



$\langle n_i \rangle = \frac{1}{2}$ on every site, one ground state

(b) $V \to \infty$

its partner, shifted by one site, same energy



$1, 0, 1, 0, \ldots$